

CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 03



By - Abhishek Sir



Recap of Previous Lecture



Topic

One-bit parity

Topic

Valid vs invalid Codewords

Topic

CRC



Topics to be Covered



Topic

CRC

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : CRC



$$\text{degree}(G(x)) = n$$
$$G(x) = x^n + \dots + 1$$

Transmitter protocol :

$$[\underbrace{M(X)} \cdot \underbrace{X^n}]$$

[Modulo-2 Division]

$$[\underbrace{G(X)}]$$



Topic : CRC



Example 1 :

$$G(X) = X^3 + X^2 + 1$$

$$M(X) = X^7 + X^4 + X^3 + X$$

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^4$$

$$[M(X) * X^3] \text{ [Modulo-2 Division] } [G(X)]$$



Topic : CRC



Modulo 2 arithmetic
[bit-wise X-OR]

$$\underline{X^3 + X^2 + 1}$$

$$X^7 + X^6 + X^5 + X^4 + X^3 + 1$$

$$\begin{array}{r} X^{10} + X^7 + X^6 + X^4 \\ \underline{X^{10} + X^9 + X^7} \end{array}$$

$$\begin{array}{r} X^9 + X^6 + X^4 \\ \underline{X^9 + X^8 + X^6} \end{array}$$

$$\begin{array}{r} X^8 + X^4 \\ \underline{X^8 + X^7 + X^5} \end{array}$$

$$\begin{array}{r} X^7 + X^5 + X^4 \\ \underline{X^7 + X^6 + X^4} \end{array}$$

$$\begin{array}{r} X^6 + X^5 \\ \underline{X^6 + X^5 + X^3} \end{array}$$

$$\begin{array}{r} X^3 \\ \underline{X^3 + X^2 + 1} \\ X^2 + 1 \end{array}$$



Topic : CRC



$$X^3 + X^2 + 1$$

Modulo 2 arithmetic
[bit-wise X-OR]

$$X^7 + X^6 + X^5 + X^4 + X^3 + 1$$

$$\begin{array}{r} X^{10} + X^7 + X^6 + X^4 \\ X^{10} + X^9 + X^7 \end{array}$$

$$\begin{array}{r} X^9 + X^6 + X^4 \\ X^9 + X^8 + X^6 \end{array}$$

$$\begin{array}{r} X^8 + X^4 \\ X^8 + X^7 + X^5 \end{array}$$

$$\begin{array}{r} X^7 + X^5 + X^4 \\ X^7 + X^6 + X^4 \end{array}$$

$$\begin{array}{r} X^6 + X^5 \\ X^6 + X^5 + X^3 \end{array}$$

$$\begin{array}{r} X^3 \\ X^3 + X^2 + 1 \end{array}$$

$$X^2 + 1$$

$$R(X) = X^2 + 1$$



Topic : Remainder Polynomial

R(X) : Remainder Polynomial function

→ n terms, [$X^{(n-1)}$ to X^0]

→ coefficients are either zero or one

CRC (Remainder) : binary string (n bits)



Topic : CRC



Example 1 :

G(X)

=

$$X^3 + X^2 + 1$$

M(X)

=

$$X^7 + X^4 + X^3 + X$$

M(X) * X³

=

$$X^{10} + X^7 + X^6 + X^4$$

[M(X) * X³] [Modulo-2 Division] [G(X)]

R(X)

=

$$1 * X^2 + 0 * X^1 + 1 * X^0$$



Topic : CRC



Transmitter protocol :

$$[M(X) * X^n] \text{ [Modulo-2 Division] } [G(X)]$$

R(X) : Remainder Polynomial function (of above equation)

Transmitter transmit : into channel, transmitted codeword

$$[M(X) * X^n] + [R(X)]$$



Topic : CRC



Example 1 :

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^4$$

$$R(X) = X^2 + 1$$

Transmitter transmit :

$$\underbrace{X^{10} + X^7 + X^6 + X^4}_{M(X) * X^3} + \underbrace{X^2 + 1}_{R(X)}$$



Topic : CRC



Example 1 :

$$\underline{G(X)} = X^3 + X^2 + 1$$

$$\underline{M(X)} = X^7 + X^4 + X^3 + X$$

$$\underline{M(X) * X^3} = X^{10} + X^7 + X^6 + X^4$$

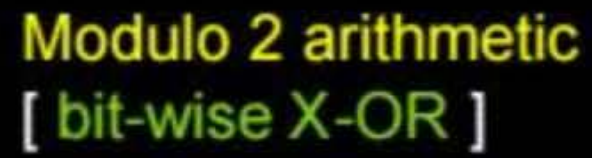
$$\underline{\text{DIVISOR}} = 1101$$

$$\underline{\text{DATA}} = 10011010$$

$$\underline{\text{DIVIDENT}} = 10011010000$$

DATA 3-zero

[$M(X) * X^3$] [Modulo-2 Division] [$G(X)$]



1 1 0 1

1 1 1 1 1 0 0 1

1 0 0 1 1 0 1 0 0 0 0

1 1 0 1

1 0 0 1

1 1 0 1

1 0 0 0

1 1 0 1

1 0 1 1

1 1 0 1

1 1 0 0

1 1 0 1

0 0 1 0

0 0 0 0

0 1 0 0

0 0 0 0

1 0 0 0

1 1 0 1

1 0 1



Topic : CRC



Modulo 2 arithmetic
[bit-wise X-OR]

$$\begin{array}{r} 1101 \overline{) 100110100000} \\ \underline{1101} \\ 1001 \\ \underline{1101} \\ 1000 \\ \underline{1101} \\ 1011 \\ \underline{1101} \\ 1100 \\ \underline{1101} \\ 1000 \\ \underline{1101} \\ 101 \end{array}$$

CRC



Topic : CRC



Example 1 :

$$G(X) = X^3 + X^2 + 1$$

$$\text{DIVISOR} = 1101$$

$$M(X) = X^7 + X^4 + X^3 + X$$

$$\text{DATA} = 10011010$$

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^4$$

$$10011010000$$

[$M(X) * X^3$] [Modulo-2 Division] [$G(X)$]

$R(X)$

$$= 1 * X^2 + 0 * X^1 + 1 * X^0$$

CRC = 101



Topic : CRC



Example 1 :

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^4$$

$$R(X) = 1 * X^2 + 0 * X^1 + 1 * X^0$$

1 0 0 1 1 0 1 0 0 0 0

CRC = 1 0 1

Transmitter transmit :

$$X^{10} + X^7 + X^6 + X^4 + X^2 + 1$$

100110101
Transmitted Codeword

1 0 0 1 1 0 1 0 1 0 1

DATA

CRC





Topic : CRC



Example 2 :

$G(X)$

=

$$X^3 + X + 1$$

$M(X)$

=

$$X^7 + X^4 + X^3 + X^2 + 1$$

$M(X) * X^3$

=

$$X^{10} + X^7 + X^6 + X^5 + X^3$$

[$M(X) * X^3$] [Modulo-2 Division] [$G(X)$]



Topic : CRC



$$X^3 + X + 1$$

Modulo 2 arithmetic
[bit-wise X-OR]

$$X^{10} + X^7 + X^6 + X^5 + X^3$$

$$X^{10} + X^8 + X^7$$

$$X^8 + X^6 + X^5 + X^3$$

$$X^8 + X^6 + X^5$$

$$X^3$$

$$X^3 + X + 1$$

$$X + 1$$

$$R(X) = X + 1$$



Topic : CRC



$$X^3 + X + 1$$

Modulo 2 arithmetic
[bit-wise X-OR]

$$X^7 + X^5 + 1$$

$$X^{10} + X^7 + X^6 + X^5 + X^3$$

$$X^{10} + X^8 + X^7$$

$$X^8 + X^6 + X^5 + X^3$$

$$X^8 + X^6 + X^5$$

$$X^3$$

$$X^3 + X + 1$$

$$X + 1$$





Topic : CRC



Example 2 :

$$G(X) = X^3 + X + 1$$

$$M(X) = X^7 + X^4 + X^3 + X^2 + 1$$

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^5 + X^3$$

$$[M(X) * X^3] \text{ [Modulo-2 Division] } [G(X)]$$

$$R(X) = 0 * X^2 + 1 * X^1 + 1 * X^0$$



Topic : CRC



Example 2 :

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^5 + X^3$$

$$R(X) = X + 1$$

Transmitter transmit :

$$\underbrace{X^{10} + X^7 + X^6 + X^5 + X^3}_{M(X) * X^3} + \underbrace{X + 1}_{R(X)}$$



Topic : CRC



Example 2 :

$$G(X) = X^3 + X + 1$$

$$M(X) = X^7 + X^4 + X^3 + X^2 + 1$$

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^5 + X^3$$

$$\text{DIVISOR} = 1011$$

$$\text{DATA} = 10011101$$

1 0 0 1 1 1 0 1 0 0 0
DATA 3 zero

[$M(X) * X^3$] [Modulo-2 Division] [$G(X)$]



Topic : CRC



Modulo 2 arithmetic
[bit-wise X-OR]

$$\begin{array}{r} 1011 \overline{) 10011101000} \\ \underline{1011} \\ 1011 \\ \underline{1011} \\ 1000 \\ \underline{1011} \\ 011 \end{array}$$



Topic : CRC



Example 2 :

$$G(X) = X^3 + X + 1$$

$$\text{DIVISOR} = 1011$$

$$M(X) = X^7 + X^4 + X^3 + X^2 + 1$$

$$\text{DATA} = 10011101$$

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^5 + X^3$$

$$10011101000$$

[$M(X) * X^3$] [Modulo-2 Division] [$G(X)$]

$R(X)$

$$= 0 * X^2 + 1 * X^1 + 1 * X^0$$

$$\text{CRC} = 011$$



Topic : CRC



Example 2 :

$$M(X) * X^3 = X^{10} + X^7 + X^6 + X^5 + X^3$$

$$R(X) = 0 * X^2 + 1 * X + 1 * X^0$$

1 0 0 1 1 1 0 1 0 0 0

CRC = 0 1 1

Transmitter transmit :

$$X^{10} + X^7 + X^6 + X^5 + X^3 + X + 1$$

1 0 0 1 1 1 0 1 0 1 1
DATA CRC

← Transmitted Codeword



Topic : CRC



Receiver protocol :

$$[M(X) * X^n + R(X)] \text{ [Modulo-2 Division] } [G(X)]$$

$R'(X)$: Remainder at receiver (of above equation)

if $R'(X) == \text{ZERO}$:
 then Receiver concluded “**No any error detected**”
else
 Receiver concluded “**Error detected**”



Topic : CRC



Example 2 :

Transmitter transmitted :

$$X^{10} + X^7 + X^6 + X^5 + X^3 + X + 1$$

1 0 0 1 1 1 0 1 0 1 1

Receiver received :

$$X^{10} + X^7 + X^6 + X^5 + X^3 + X + 1$$

1 0 0 1 1 1 0 1 0 1 1

$$G(X) = X^3 + X + 1$$

$$\text{DIVISOR} = 1011$$



Topic : CRC



Modulo 2 division
[bit-wise X-OR]

$$\begin{array}{r} 1011 \overline{) 10011101011} \\ \underline{1011} \\ 1011 \\ \underline{1011} \\ 1011 \\ \underline{1011} \\ 000 \end{array}$$



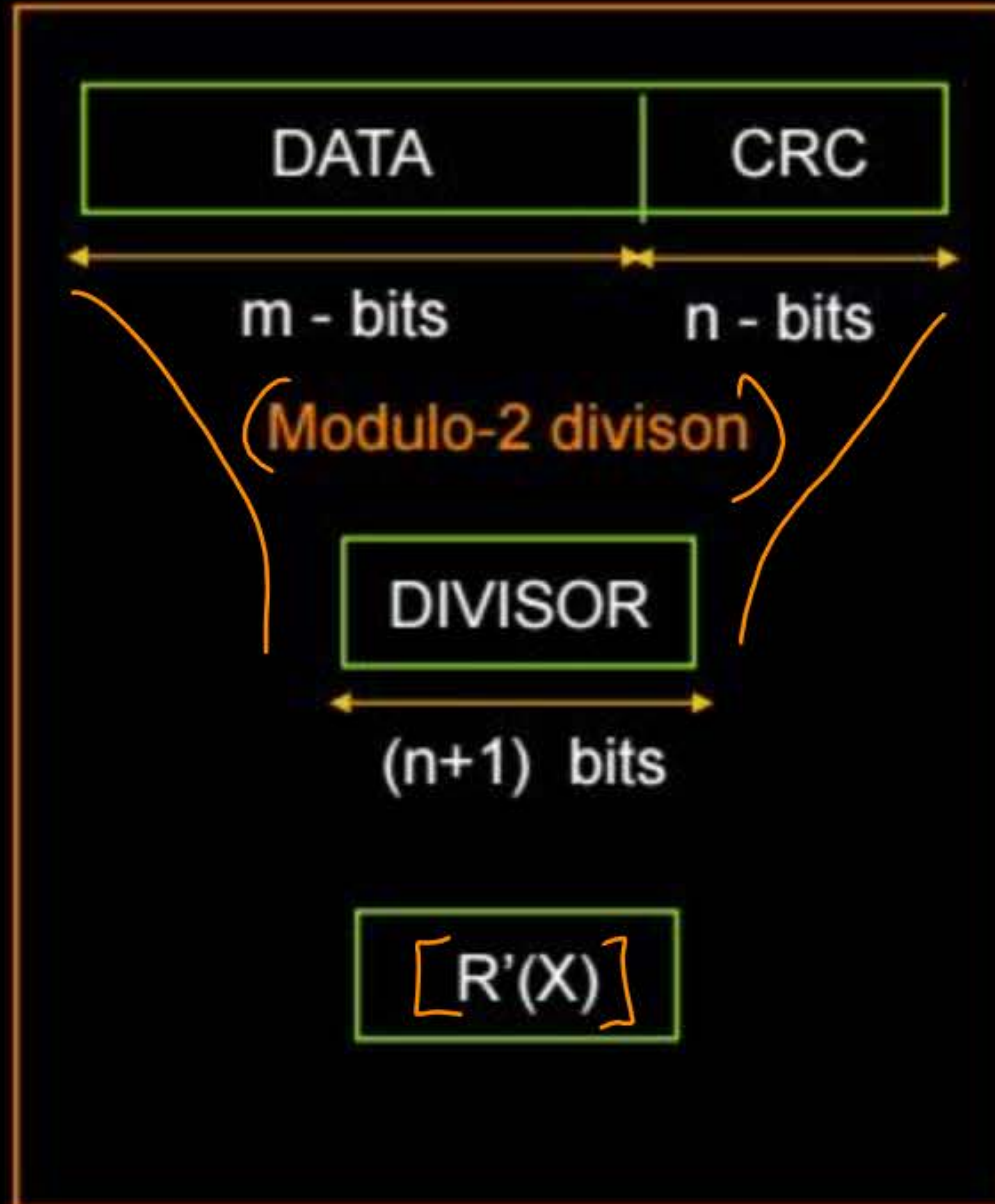
Topic : CRC



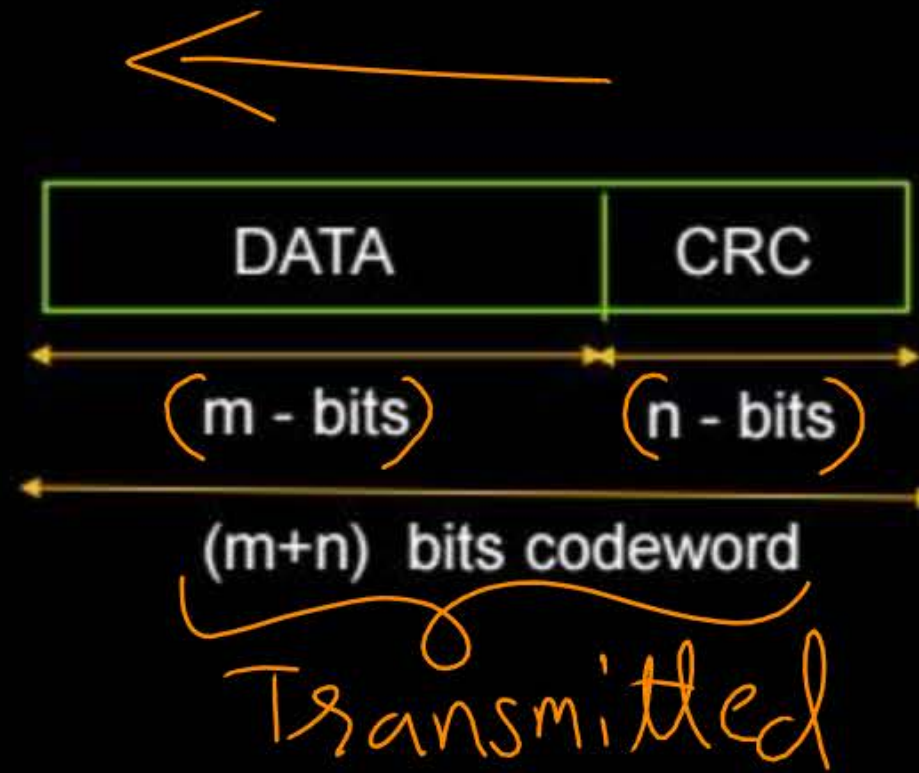
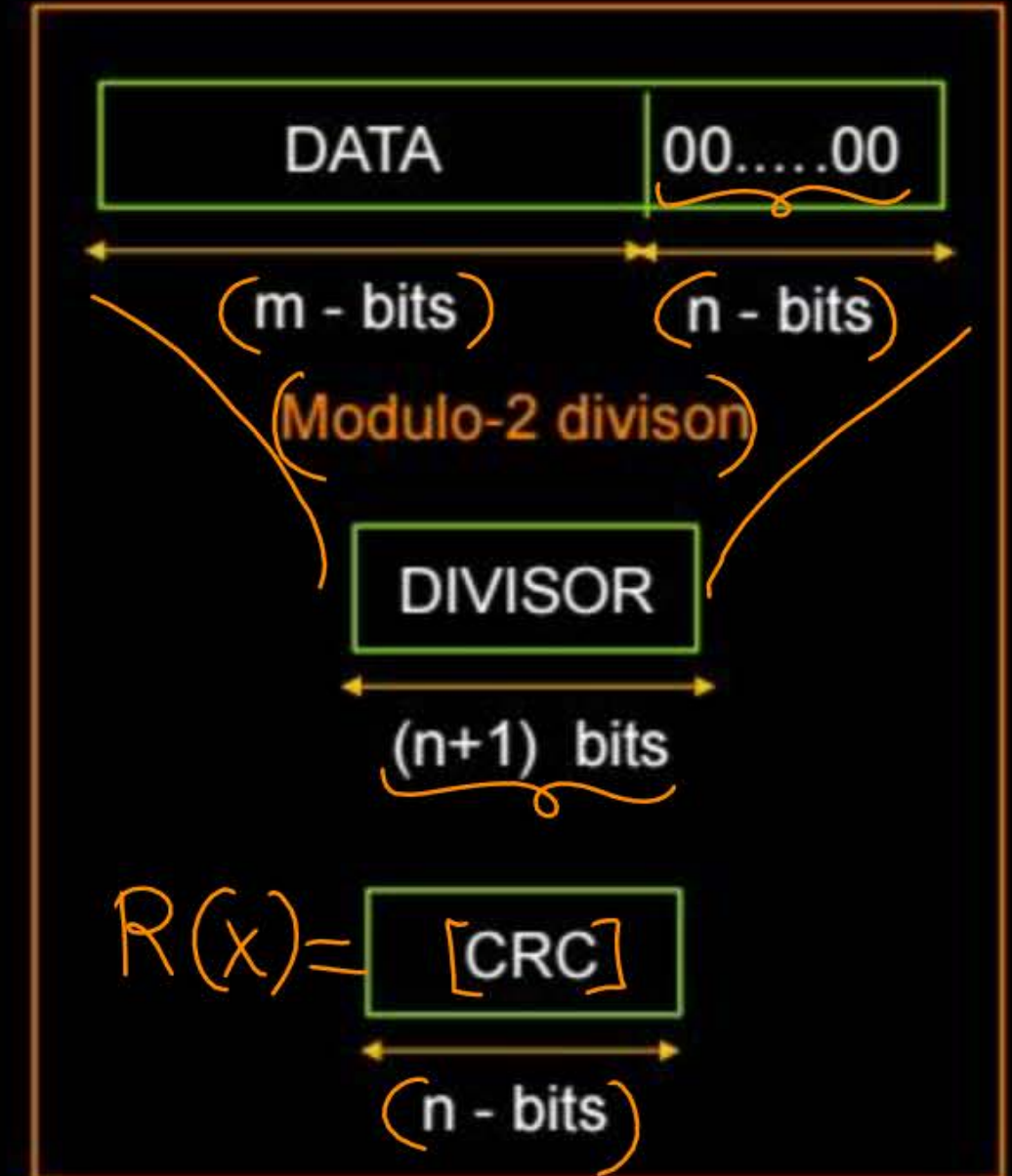
$$G(X) = X^n + \dots + 1 \quad \text{where } n > 0$$

$\text{degree}[G(X)] = n$

Receiver



Sender (Transmitter)



#Q. Consider the message $M = 1010001101$. The cyclic redundancy check (CRC) for this message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is

- (A) 01110
- (B) 01011
- (C) 10101
- (D) 10110

[GATE 2005]
 IIT-B
 (H.W.)

#Q. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is :

- (A) 11001001000
- (B) 11001001011
- (C) 11001010
- (D) 110010010011

[GATE 2007]
 IIT-K
 H.W.

#Q. A computer network uses polynomials over $GF(2)$ for error checking with 8 bits as information bits and uses $x^3 + x + 1$ as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as :

- (A) 01011011010
- (B) 01011011011
- (C) 01011011101
- (D) 01011011100

[GATE 2017]

IIT-R
H.W.

#Q. Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial X^3+X+1 . Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the checkbit sequence $c_2c_1c_0$ is :

- (A) 101
- (B) 110
- (C) 100
- (D) 111

[GATE 2021, Set-2, 2-Mark]

IIT-B, H.W.



2 mins Summary



Topic

CRC



THANK - YOU

